MIL-P-63142A (AR)
AMENDMENT 5
20 May 1983
SUPERSEDING
AMENDMENT 4
17 January 1983

MILITARY SPECIFICATION

PROJECTILE, 8 INCH, HE, RA, M650
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-P-63142A (AR), dated 08 February, 1981, and is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

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* 3.6.1.2.2, add the following: "unless the rocket motor fails to ignite."

* Delete paragraph 3.6.2 in its entirety and substitute the following:

"3.6.2 Safety. There shall be no premature burst or indication of metal parts separation in the gun bore, or in flight, when fired at any of the conditions specified in 4.4.3.4. In addition, no projectile fired in the safety phase shall have a horizontal range less than 97.5 percent of the mean unless the rocket motor fails to ignite."

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4.4.2.1, change title to read: "Delay Assy and Motor Body prior to Assembly."

Add the following Major defect:

Major category:

"Major 103: Rocket motor body protective cap missing or improper 100% 3.2 Visual."

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4.4.2.5, delete "Critical 1: Nozzle insert missing, loose, inverted, or damaged" and substitute the following:

"Critical 1: Nozzle insert missing, loose, or damaged."

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4.4.2.8, add the following Major defects:

Major category:

"Major 102: Forward protective cap missing or improper 100% 3.2 Visual.

Major 103: Aft protective cap missing or improper 100% 3.2 Visual"

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4.4.2.9, Major category:

Major 102, Inspection Method column: Delete "Gage" and substitute "Visual".

Major 104, Inspection Method column: Delete: "Visual" and substitute "Gage".

Add the following Major defects:

Major category:

"Major 107: Pellet cavity depth incorrect 0.40% 3.2 Gage.

Major 108: Pellet cavity diameter incorrect 0.40% 3.2 Gage."

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4.4.2.14, Critical defect 1: Delete "O-ring groove".

Major defect 106: Delete in its entirety.

Add the following Major defects:

Major category:

"Major 106: Forward grain cushion improper, missing or improperly lubricated, prior to application of sealant 0.40% 3.2 Visual

Major 108: Aft protective cap missing or improper on aft warhead 100% 3.2 Visual."

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4.4.2.15, add the following Minor defect:

Minor category:

"Minor 205: Lifting plug improperly lubricated 0.65% 3.2 Visual."

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4.4.2.16, Major defect 103:

Delete in its entirety and substitute the following:

Major category:

"Major 103: Strapping or staples missing or loose 0.40% 3.2 Visual."

Minor Defect 203:

Delete in its entirety and substitute the following:

Minor category:

"Minor 203: Strapping or staples improperly engaged 0.65% 3.2 Visual".

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4.4.3.2, add the following:

"As an alternate to the once an hour torque check sample, a verification check may be made on the torque assembly machine. In this case, the torque readout from the assembly machine shall be verified for transducer accuracy at the beginning and end of the operating shift and after each four hour production interval. If the torque assembly machine fails the verification test, all the projectiles assembled during that production interval shall be rejected. In the event of two consecutive gap rejects, the torque assembly machine shall be checked for proper calibration and operation at that time. Failure of these checks will reject those projectiles assembled since the last acceptable verification test. If the torque display mechanism indicates a torque not within the specified range, the torque assembly machine shall automatically shut down and require appropriate corrections to restore its operation."

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* 4.4.3.4, Critical defect 2, add the following: "unless the rocket motor fails to ignite."

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Delete Critical defect 3 in its entirety.

Add Major defect 102 as follows:

Major category:

"Major 102: Rocket motor fails to ignite."

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* Add paragraph 4.4.3.4.2.3 as follows:

"4.4.3.4.2.3 Failure of a rocket motor to ignite during +145°F safety tests shall be considered a Major defect, and the lot shal be rejected. If a motor ignition failure occurs during -50°F safety tests, the lot shall be referred to the procuring agency for disposition (see 6.14)."

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4.4.3.7, add the following:

"As an alternate to the once an hour torque check sample, a verification check may be made on the torque assembly machine. In this case, the torque readout from the assembly machine shall be verified for transducer accuracy at the beginning and end of the operating shift and after each four hour production interval. If the torque assembly machine fails the verification test, all the projectiles assembled during that production interval shall be rejected. In the event of two consecutive gap rejects, the torque assembly machine shall be checked for proper calibration and operation at that time. Failure of these checks will reject those projectiles assembled since the last acceptable verification test. If the torque display mechanism indicates a torque not within the specified range, the torque assembly machine shall automatically shut down and require appropriate corrections to restore its operation."

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* 4.5.3.2: Change "-65°F" to read "-50°F", (two places).

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* Add paragraph 6.14 as follows:

"6.14 Rocket motor ignition failure at -50°F. Degradation of rocket motor ignition reliability is expected at temperatures below -25°F. When an ignition failure occurs at -50°F during acceptance testing, the procuring agency must evaluate the following prior to determining lot disposition:

a. The delay assembly shall have functioned properly as evidenced by review of radar data.

b. The failure shall not be related to a change in components and/or processes.

c. The effect on compatibility with the M753 NUC round must be considered. This may require a retest at -30°F.

d. The rate of failure should not be significantly different from prior experience."

The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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